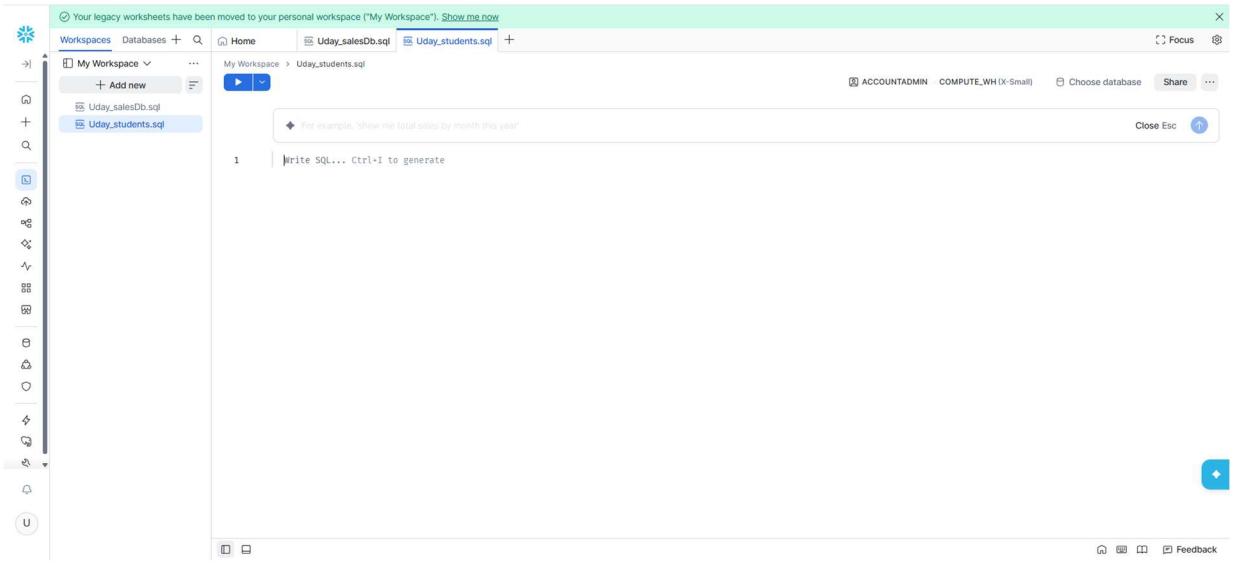
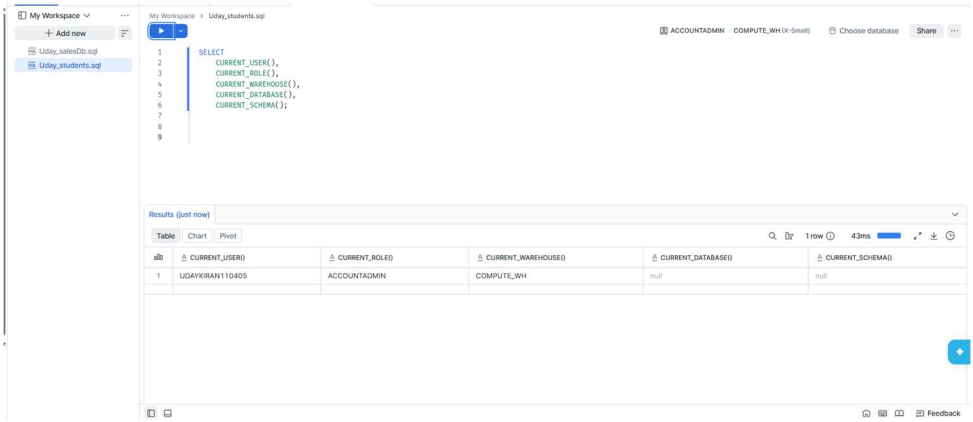
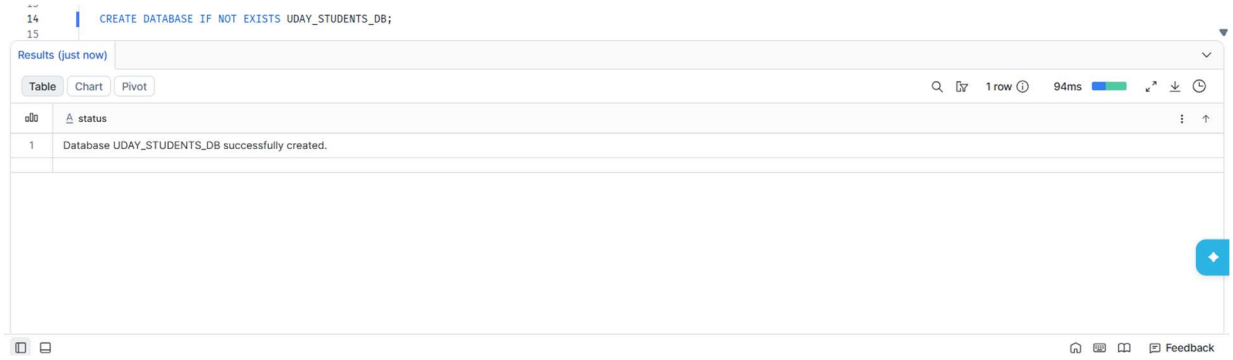
Creating the workspace:



Check the current Snowflake session



Creating the database



Checking :

```
16  SHOW DATABASES;
17
```

Results (just now)

Table | Chart | Pivot — 6 rows — 62ms

	created_on	name	is_default	is_current	origin	owner	comment	options	
1	2026-11-02 01:31:27.651 -0800	SNOWFLAKE	N	N	SFCOGSOPS.SNOWHOUSE_LOCAL_AWSAPSOUTHEAST7.A				0
2	2026-11-02 01:31:31.922 -0800	SNOWFLAKE_LEARN	N	N		ACCOUNTADMIN	Created by Snowflake		1
3	2026-11-02 01:31:31.474 -0800	SNOWFLAKE_SAMP	N	N	SFSALESSHARED.SFC_SAMPLES_AWSAPSOUTHEAST7.SAI	ACCOUNTADMIN	Preloaded TPCH Data		0
4	2026-11-02 02:01:36.009 -0800	UDAY_RETAILS_DB	N	N		SYSADMIN			1
5	2026-11-02 03:25:25.989 -0800	UDAY_STUDENTS_D	N	Y		ACCOUNTADMIN			
6	2026-11-02 01:33:11.526 -0800	USER$UDAYKIRAN1	N	N					

Using The Like

```
18  SHOW DATABASES LIKE 'UDAY_STUDENTS_DB';
```

Results (just now)

Table | Chart | Pivot — 1 row — 51ms

	created_on	name	is_default	is_current	origin	owner	comment	options	retention_time	kind
1	2026-11-02 03:25:25	UDAY_STUDENTS_DB	N	Y		ACCOUNTADMIN			1	STANDARD

Creating the Schema :

```
20  CREATE SCHEMA IF NOT EXISTS UDAY_STUDENTS_DB.UDAY_STUDENTS_RAW;
21
```

Results (just now)

Table | Chart | Pivot — 1 row — 45ms

	status
1	Schema UDAY_STUDENTS_RAW successfully created.

Checking :

```
22  SHOW SCHEMAS IN DATABASE UDAY_STUDENTS_DB;
```

Results (just now)

Table | Chart | Pivot — 3 rows — 56ms

	created_on	name	is_default	is_current	database_name	owner	comment	options
1	2026-11-02 03:26:53.058 -0800	INFORMATION_SCHEMA	N	N	UDAY_STUDENTS_DB		Views describing the contents of schemas in this database	
2	2026-11-02 03:25:26.012 -0800	PUBLIC	N	N	UDAY_STUDENTS_DB	ACCOUNTADMIN		
3	2026-11-02 03:26:33.753 -0800	UDAY_STUDENTS_RAW	N	Y	UDAY_STUDENTS_DB	ACCOUNTADMIN		

Creating the Virtual Warehouse :

```
CREATE WAREHOUSE IF NOT EXISTS UDAY_STUDENTS_WH
WITH
    WAREHOUSE_SIZE = 'XSMALL'
    AUTO_SUSPEND = 60
    AUTO_RESUME = TRUE
    INITIALLY_SUSPENDED = TRUE;
```

Results (just now) — Table | Chart | Pivot — 1 row — 85ms

	status
1	Warehouse UDAY_STUDENTS_WH successfully created.

Checking :

```
SHOW WAREHOUSES LIKE 'UDAY_STUDENTS_WH';
```

Results (just now) — Table | Chart | Pivot — 1 row — 54ms

	name	state	type	size	running	queued	is_default	is_current	auto_suspend	auto_res
1	UDAY_STUDENTS_WH	SUSPENDED	STANDARD	X-Small	0	0	N	Y	60	true

Select our database, schema, and warehouse:

```
USE DATABASE UDAY_STUDENTS_DB;
 Ctrl+I to generate
USE SCHEMA UDAY_STUDENTS_RAW;

USE WAREHOUSE UDAY_STUDENTS_WH;
```

Results (just now) — Table | Chart | Pivot — 1 row — 23ms

	status
1	Statement executed successfully.

Checking all the Db, Schema and Warehouse :

```
42  SELECT
43      CURRENT_USER(),
44      CURRENT_ROLE(),
45      CURRENT_WAREHOUSE(),
46      CURRENT_DATABASE(),
47      CURRENT_SCHEMA();
```

Results (just now)

Table | Chart | Pivot — 1 row — 24ms

	CURRENT_USER()	CURRENT_ROLE()	CURRENT_WAREHOUSE()	CURRENT_DATABASE()	CURRENT_SCHEMA()
1	UDAYKIRAN110405	ACCOUNTADMIN	UDAY_STUDENTS_WH	UDAY_STUDENTS_DB	UDAY_STUDENTS_RAW

Creating the csv Files :

1. students.csv(Contains the Spaces and Null Values)

dents.csv | orders.csv | sales.csv | customers.c | Uday_studer | enrollments. | Untitled | courses.csv | stude

File Edit View

```
STUDENT_ID,STUDENT_NAME,AGE,DEPARTMENT,CGPA,CITY
101,Uday Kiran,21,CSE,9.75,Hyderabad
102,Rahul,22,ECE,8.45,Khammam
103,Ananya,20,CSE,9.20,Warangal
104,Sneha,21,IT,8.90,hyderabad
105,Arjun,22,ECE,7.85,
106,Priya,20,IT,9.50, WARANGAL
107, ,21,CSE,8.75,Hyderabad
108,Meghana,,AI ML,9.10,Khammam
109,Varun,20, ,8.60,hyderabad
110,Divya,21,IT,9.30,Warangal
111,Kiran,22,CSE,,Khammam
112, Rakesh ,21,ECE,8.25, Hyderabad
113,Neha,20,IT,9.15,
114, NULL ,22,CSE,8.80,Warangal
115,Manoj, ,AI ML,9.00,Khammam
```

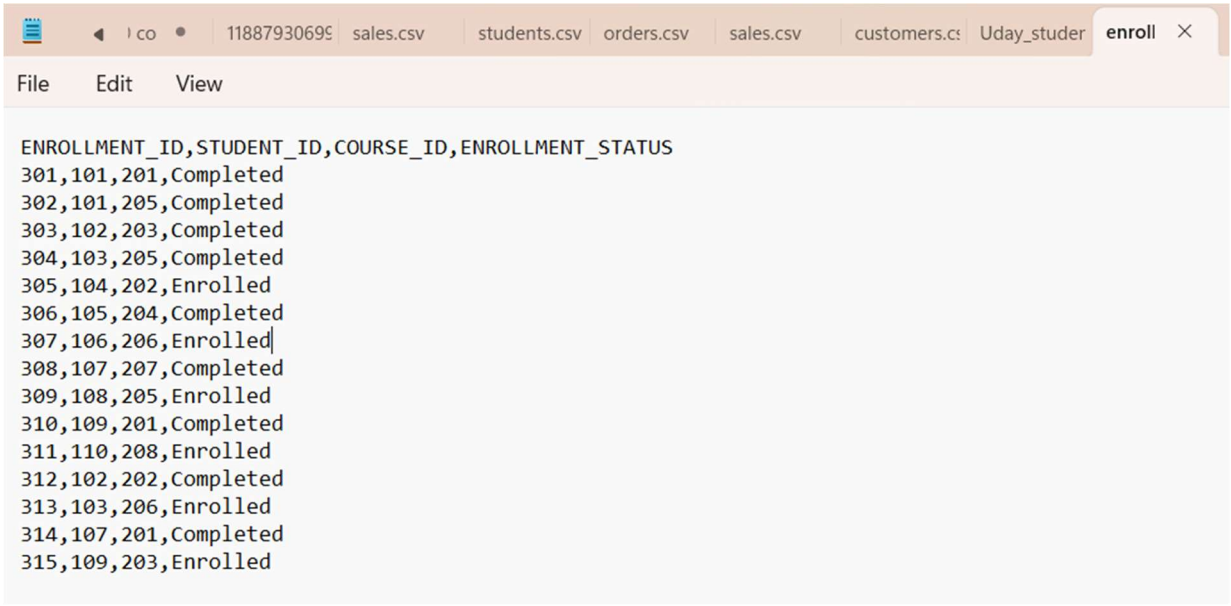
2.courses.csv

dents.csv | orders.csv | sales.csv | customers.c | Uday_studer | enrollments. | Untitled | cours | students.csv

File Edit View

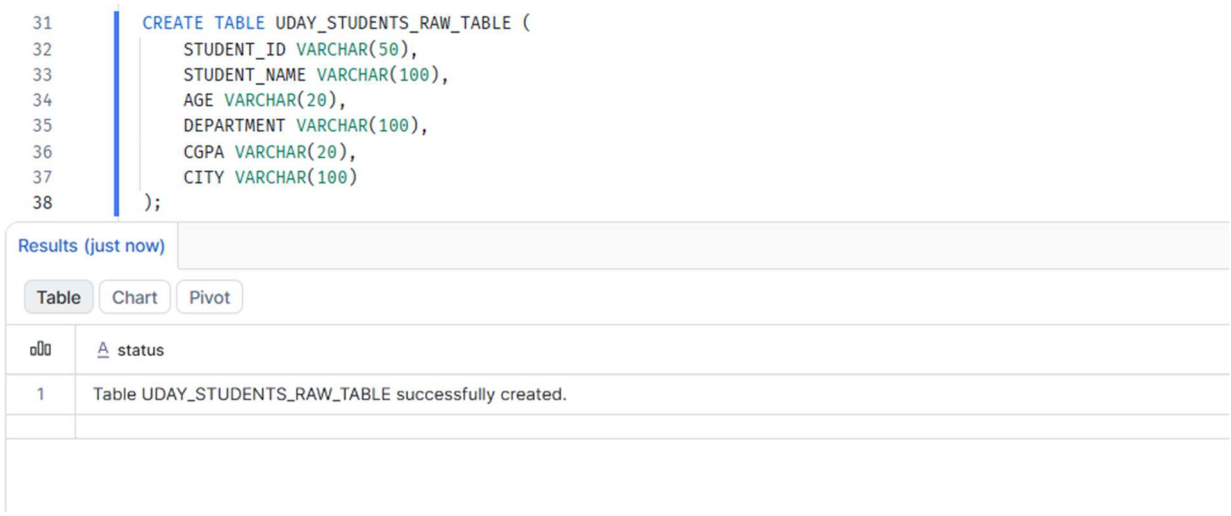
```
ENROLLMENT_ID,STUDENT_ID,COURSE_ID,ENROLLMENT_STATUS
301,101,201,Completed
302,101,205,Completed
303,102,203,Completed
304,103,205,Completed
305,104,202,Enrolled
306,105,204,Completed
307,106,206,Enrolled
308,107,207,Completed
309,108,205,Enrolled
310,109,201,Completed
311,110,208,Enrolled
312,102,202,Completed
313,103,206,Enrolled
314,107,201,Completed
315,109,203,Enrolled
```

3.enrollments.csv



Create the tables:

Students RAW table



Courses RAW table

```
CREATE TABLE UDAY_COURSES_RAW_TABLE (
    COURSE_ID VARCHAR(50),
    COURSE_NAME VARCHAR(100),
    CATEGORY VARCHAR(100),
    COURSE_FEE VARCHAR(50)
);
```

Results (just now)

Table | Chart | Pivot — 1 row — 118ms

	status
1	Table UDAY_COURSES_RAW_TABLE successfully created.

Enrollments RAW table

```
CREATE TABLE UDAY_ENROLLMENTS_RAW_TABLE (
    ENROLLMENT_ID VARCHAR(50),
    STUDENT_ID VARCHAR(50),
    COURSE_ID VARCHAR(50),
    ENROLLMENT_STATUS VARCHAR(50)
);
```

Results (just now)

Table | Chart | Pivot — 1 row — 93ms

	status
1	Table UDAY_ENROLLMENTS_RAW_TABLE successfully created.

Checking :

```
SHOW TABLES;
```

Results (just now)

Table | Chart | Pivot — 4 rows — 50ms

	created_on	name	database_name	schema_name	kind	comment	cluster_by
1	2026-11-04 04:42:07.083 -0800	UDAY_COURSES_RAW_TABLE	UDAY_STUDENTS_DB	UDAY_STUDENTS_RAW	TABLE		
2	2026-11-04 04:42:42.676 -0800	UDAY_ENROLLMENTS_RAW_TABLE	UDAY_STUDENTS_DB	UDAY_STUDENTS_RAW	TABLE		
3	2026-11-02 03:29:00.787 -0800	UDAY_STUDENTS	UDAY_STUDENTS_DB	UDAY_STUDENTS_RAW	TABLE		
4	2026-11-04 04:38:46.535 -0800	UDAY_STUDENTS_RAW_TABLE	UDAY_STUDENTS_DB	UDAY_STUDENTS_RAW	TABLE		

Create the CSV file format

```
CREATE FILE FORMAT IF NOT EXISTS UDAY_STUDENTS_CSV
TYPE = CSV
FIELD_DELIMITER = ','
SKIP_HEADER = 1
FIELD_OPTIONALLY_ENCLOSED_BY = '"'
EMPTY_FIELD_AS_NULL = TRUE;
```

Results (just now)

Table | Chart | Pivot

	status
1	UDAY_STUDENTS_CSV already exists, statement succeeded.

Checking :

```
SHOW FILE FORMATS IN UDAY_STUDENTS_DB.UDAY_STUDENTS_RAW;
```

Results (just now) — Table | Chart | Pivot — 1 row — 42ms

	created_on	name	database_name	schema_name	type	owner	comment	format_options	owner_role_type
1	2026-11-02 03:30:26.4	UDAY_STUDENTS_CSV	UDAY_STUDENTS_DB	UDAY_STUDENTS_RAW	CSV	ACCOUNTADMIN		{"TYPE":"CSV","RECORD	ROLE

```
DESCRIBE FILE FORMAT UDAY_STUDENTS_CSV;
```

Results (just now) — Table | Chart | Pivot — 24 rows — 34ms

	property	property_type	property_value	property_default
	BINARY_FORMAT 4.2% COMPRESSION 4.2% +22 more	String 54.2% Boolean 37.5% +2 more	true 20.8% AUTO 16.7% +12 more	true 20.8% AUTO 16.7% +11 more
1	TYPE	String	CSV	CSV
2	RECORD_DELIMITER	String	\n	\n
3	FIELD_DELIMITER	String	,	,
4	FILE_EXTENSION	String		
5	SKIP_HEADER	Integer	1	0
6	PARSE_HEADER	Boolean	false	false

Create two separate stages

Stage 1 – Students CSV

```
68  CREATE STAGE IF NOT EXISTS UDAY_STUDENTS_STAGE
69  FILE_FORMAT = UDAY_STUDENTS_CSV;
```

Results (just now)

Table | Chart | Pivot

	status
1	UDAY_STUDENTS_STAGE already exists, statement succeeded.

Stage 2 – Courses and Enrollments CSVs

```
72  CREATE STAGE IF NOT EXISTS UDAY_ACADEMIC_STAGE
73  FILE_FORMAT = UDAY_STUDENTS_CSV;
74
```

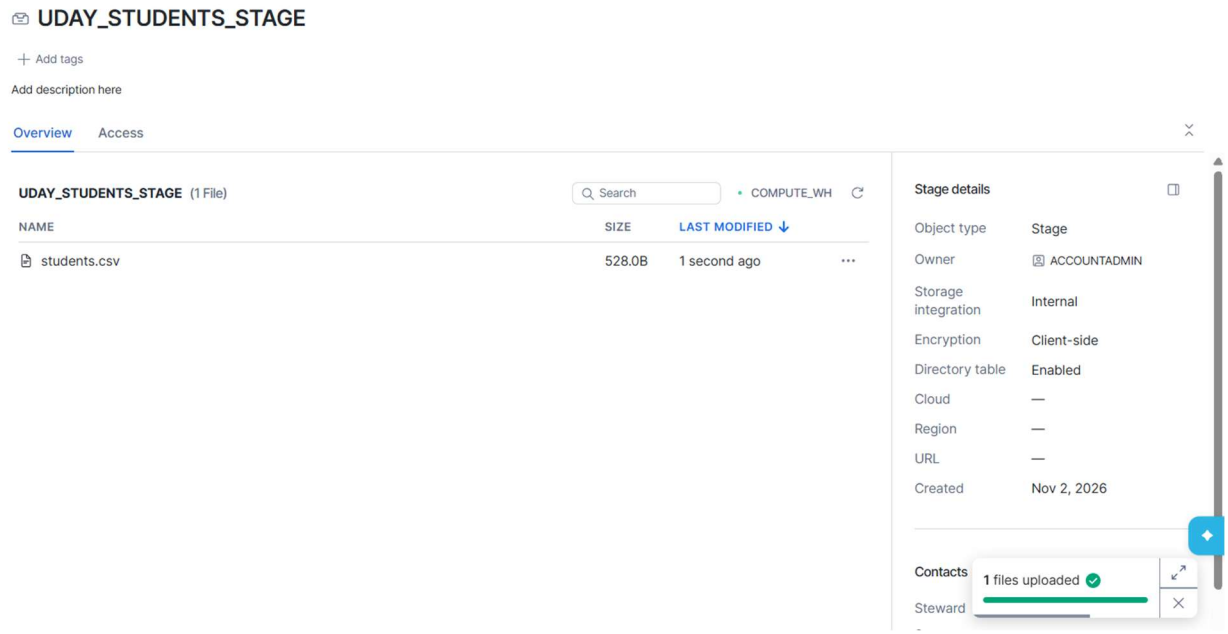
Results (just now)

Table | Chart | Pivot — 1 row

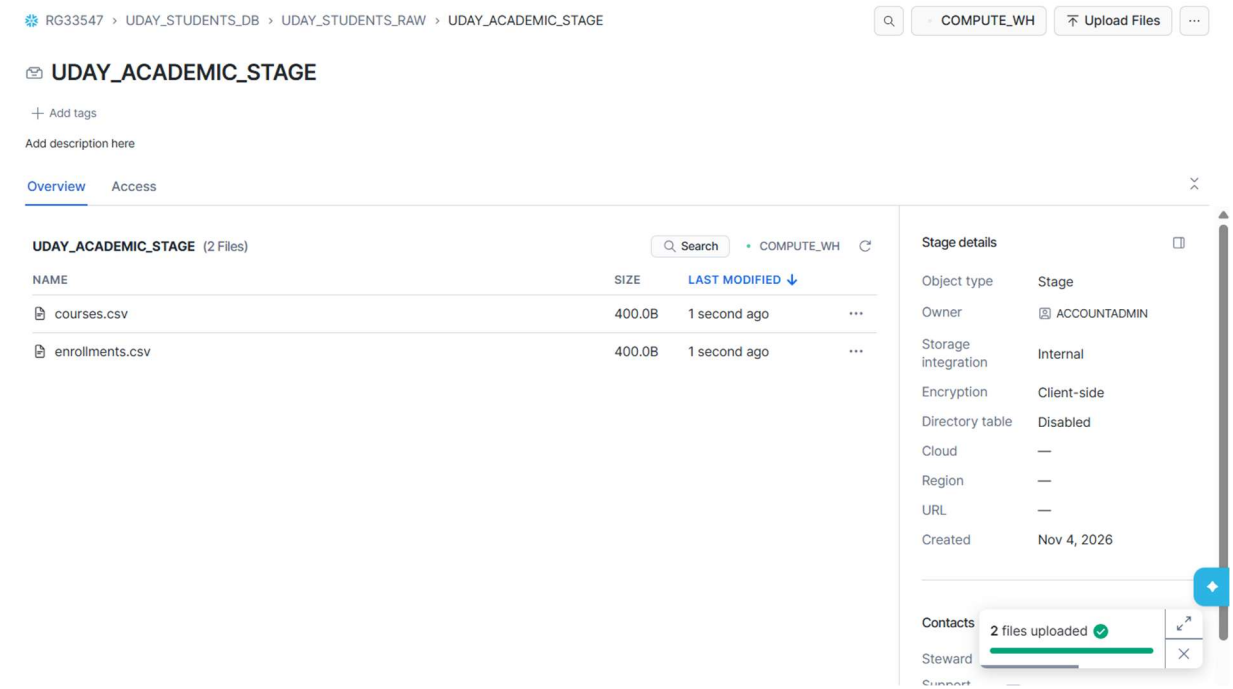
	status
1	Stage area UDAY_ACADEMIC_STAGE successfully created.

Upload CSV files to the stages

On Students Stage :

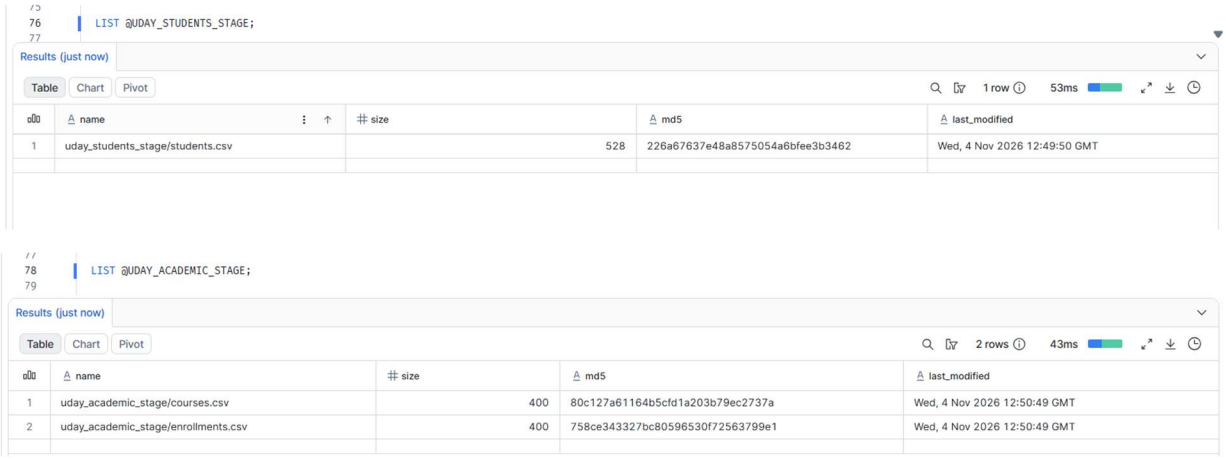


Uploading the Multiple Files in the Same stage :



->2 Files Can be Uploaded into the same stage as they are in the same csv Format

Checking :



Load the CSV files into their respective RAW tables

Load students

```
COPY INTO UDAY_STUDENTS_RAW_TABLE
FROM @UDAY_STUDENTS_STAGE
FILE_FORMAT = (FORMAT_NAME = UDAY_STUDENTS_CSV)
ON_ERROR = 'ABORT_STATEMENT';
```

Results (just now) — Table | Chart | Pivot — 1 row — 646ms

	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_error_line	first_error_character	first_er
1	uday_students_stage/s	LOADED	15	15	1	0	null	null	null	null

Load courses

```
COPY INTO UDAY_COURSES_RAW_TABLE
FROM @UDAY_ACADEMIC_STAGE/courses.csv
FILE_FORMAT = (FORMAT_NAME = UDAY_STUDENTS_CSV)
ON_ERROR = 'ABORT_STATEMENT';
```

Results (just now) — Table | Chart | Pivot — 1 row — 789ms

	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_error_line	first_error_character	first_er
1	uday_academic_stage/	LOADED	15	15	1	0	null	null	null	null

Load enrollments

```
COPY INTO UDAY_ENROLLMENTS_RAW_TABLE
FROM @UDAY_ACADEMIC_STAGE/enrollments.csv
FILE_FORMAT = (FORMAT_NAME = UDAY_STUDENTS_CSV)
ON_ERROR = 'ABORT_STATEMENT';
```

Results (just now) — Table | Chart | Pivot — 1 row — 570ms

	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_error_line	first_error_character	first_er
1	uday_academic_stage/	LOADED	15	15	1	0	null	null	null	null

Checking the Loaded Records:

```
SELECT * FROM UDAY_STUDENTS_RAW_TABLE;
```

Results (just now) — Table | Chart | Pivot — 15 rows — 268ms

	STUDENT_ID	STUDENT_NAME	AGE	DEPARTMENT	CGPA	CITY
	101 6.7% 102 6.7% +13 more	6.7% NULL 6.7% +13 more	21 33.3% 20 26.7% +2 more	CSE 33.3% IT 26.7% +3 more	7.85 6.7% 8.25 6.7% +12 more	Khammam 26.7% Warangal 20.0% +5 more
1	101	Uday Kiran	21	CSE	9.75	Hyderabad
2	102	Rahul	22	ECE	8.45	Khammam
3	103	Ananya	20	CSE	9.20	Warangal
4	104	Sneha	21	IT	8.90	hyderabad
5	105	Arjun	22	ECE	7.85	null
6	106	Priya	20	IT	9.50	WARANGAL

```
102
103  SELECT * FROM UDAY_COURSES_RAW_TABLE;
```

Results (just now) — Table | Chart | Pivot — 15 rows — 128ms

	COURSE_ID	COURSE_NAME	CATEGORY	COURSE_FEE
	301 6.7% 302 6.7% +13 more	101 13.3% 102 13.3% +8 more	201 20.0% 205 20.0% +6 more	Completed 60.0% Enrolled 40.0%
1	301	101	201	Completed
2	302	101	205	Completed
3	303	102	203	Completed
4	304	103	205	Completed
5	305	104	202	Enrolled
6	306	105	204	Completed

```
104
105  SELECT * FROM UDAY_ENROLLMENTS_RAW_TABLE;
```

Results (just now) — Table | Chart | Pivot — 15 rows — 88ms

	ENROLLMENT_ID	STUDENT_ID	COURSE_ID	ENROLLMENT_STATUS
	301 6.7% 302 6.7% +13 more	101 13.3% 102 13.3% +8 more	201 20.0% 205 20.0% +6 more	Completed 60.0% Enrolled 40.0%
1	301	101	201	Completed
2	302	101	205	Completed
3	303	102	203	Completed
4	304	103	205	Completed
5	305	104	202	Enrolled
6	306	105	204	Completed

->Students have the “ ”(Space) as the Empty Field and ‘’(Empty Field) as null

Checking the Null Values :

```
116  SELECT
117      STUDENT_ID,
118      STUDENT_NAME,
119      AGE,
120      DEPARTMENT,
121      CGPA,
122      CITY
123  FROM UDAY_STUDENTS_RAW_TABLE
124  WHERE
125      NULLIF(TRIM(STUDENT_NAME), '') IS NULL
126      OR NULLIF(TRIM(AGE), '') IS NULL
127      OR NULLIF(TRIM(DEPARTMENT), '') IS NULL
128      OR NULLIF(TRIM(CGPA), '') IS NULL
129      OR NULLIF(TRIM(CITY), '') IS NULL;
```

Results (just now) — Table | Chart | Pivot — 7 rows — 314ms

	STUDENT_ID	STUDENT_NAME	AGE	DEPARTMENT	CGPA	CITY
1	105	Arjun	22	ECE	7.85	null
2	107		21	CSE	8.75	Hyderabad
3	108	Meghana	null	AI ML	9.10	Khammam
4	109	Varun	20		8.60	hyderabad
5	111	Kiran	22	CSE	null	Khammam
6	113	Neha	20	IT	9.15	
7	115	Manoj		AI ML	9.00	Khammam

NULLIF() Function:

Syntax :

NULLIF(expression1, expression2)

It works like this:

- If both expressions are equal, it returns NULL.
- If both expressions are different, it returns the first expression.
- Step 1: `TRIM(STUDENT_NAME)`

Original value	After TRIM
Uday	Uday
Preety	Preety
''	''
' '	''
NULL	NULL

- Step 2: `NULLIF(TRIM(STUDENT_NAME), '')`

Value after TRIM	NULLIF output
Uday	Uday
Preety	Preety
''	NULL
''	NULL
NULL	NULL

```
SELECT
   STUDENT_ID,
   STUDENT_NAME,
   AGE,
   DEPARTMENT,
   CGPA,
   CITY
FROM UDAY_STUDENTS_RAW_TABLE
WHERE
   NULLIF(TRIM(STUDENT_NAME), '') IS NULL
   OR NULLIF(TRIM(AGE), '') IS NULL
   OR NULLIF(TRIM(DEPARTMENT), '') IS NULL
   OR NULLIF(TRIM(CGPA), '') IS NULL
   OR NULLIF(TRIM(CITY), '') IS NULL;
```

It retrieves student records where at least one of the specified columns contains:
An actual NULL value.
An empty string ('').
A string containing only spaces (' ').

Number Transforming Functions:

TRY_TO_NUMBER() Converts text into numbers safely.

Converts a text value into a number. If conversion is not possible, it returns NULL instead of throwing an error.

Ex:

```
SELECT
   TRY_TO_NUMBER('21') AS AGE1,
   TRY_TO_NUMBER('25') AS AGE2,
   TRY_TO_NUMBER('Twenty') AS AGE3;
```

AGE1 AGE2 AGE3

21 25 NULL

TRY_TO_DECIMAL() Converts CGPA into decimal format safely.

Converts a value into a decimal number while allowing you to specify precision and scale.

Syntax:

TRY_TO_DECIMAL(expression, precision, scale)

Two important concepts:

- Precision: Maximum total number of digits.
- Scale: Maximum number of digits after the decimal point.

```
SELECT
   TRY_TO_DECIMAL('9.75', 5, 2) AS CGPA1,
   TRY_TO_DECIMAL('8.50', 5, 2) AS CGPA2,
   TRY_TO_DECIMAL('Excellent', 5, 2) AS CGPA3;
```

Output:

CGPA1	**CGPA2**	**CGPA3**
9.75	8.50	NULL

Transform Students

My Workspace > Uday_Students.sql

```
131
132  CREATE OR REPLACE TABLE UDAY_STUDENTS AS
133  SELECT
134      TRY_TO_NUMBER(TRIM(STUDENT_ID)) AS STUDENT_ID,
135
136      COALESCE(
137          NULLIF(UPPER(TRIM(STUDENT_NAME)), ''),
138          'UNKNOWN'
139      ) AS STUDENT_NAME,
140
141      TRY_TO_NUMBER(NULLIF(TRIM(AGE), '')) AS AGE,
142
143      COALESCE(
144          NULLIF(UPPER(TRIM(DEPARTMENT)), ''),
145          'UNKNOWN'
146      ) AS DEPARTMENT,
147
148      TRY_TO_DECIMAL(
149          NULLIF(TRIM(CGPA), ''),
150          3, 2
151      ) AS CGPA,
152
153      COALESCE(
154          NULLIF(UPPER(TRIM(CITY)), ''),
155          'UNKNOWN'
156      ) AS CITY
157
158  FROM UDAY_STUDENTS_RAW_TABLE;
```

Results (just now)

Table | Chart | Pivot

	status
1	Table UDAY_STUDENTS successfully created.

Transform Enrollments

```
161  CREATE OR REPLACE TABLE UDAY_ENROLLMENTS AS
162  SELECT
163      TRY_TO_NUMBER(TRIM(ENROLLMENT_ID)) AS ENROLLMENT_ID,
164
165      TRY_TO_NUMBER(TRIM(STUDENT_ID)) AS STUDENT_ID,
166
167      TRY_TO_NUMBER(TRIM(COURSE_ID)) AS COURSE_ID,
168
169      COALESCE(
170          NULLIF(UPPER(TRIM(ENROLLMENT_STATUS)), ''),
171          'UNKNOWN'
172      ) AS ENROLLMENT_STATUS
173
174  FROM UDAY_ENROLLMENTS_RAW_TABLE;
```

Results (just now)

Table | Chart | Pivot — 1 row — 795ms

	status
1	Table UDAY_ENROLLMENTS successfully created.

Transform Courses

```
178  CREATE OR REPLACE TABLE UDAY_COURSES AS
179  SELECT
180      TRY_TO_NUMBER(TRIM(COURSE_ID)) AS COURSE_ID,
181
182      COALESCE(
183          NULLIF(UPPER(TRIM(COURSE_NAME)), ''),
184          'UNKNOWN'
185      ) AS COURSE_NAME,
186
187      COALESCE(
188          NULLIF(UPPER(TRIM(CATEGORY)), ''),
189          'UNKNOWN'
190      ) AS CATEGORY,
191
192      COALESCE(NULLIF(TRIM(COURSE_FEE), ''),'UNKNOWN') AS COURSE_FEE
193
194  FROM UDAY_COURSES_RAW_TABLE;
```

Results (just now)

Table | Chart | Pivot — 1 row — 611ms

	status
1	Table UDAY_COURSES successfully created.

Checking :

```
196
197  SELECT * FROM UDAY_STUDENTS;
198
```

Results (just now)

Table | Chart | Pivot | Next steps — 15 rows — 123ms

	STUDENT_ID	STUDENT_NAME	AGE	DEPARTMENT	CGPA	CITY
	101 – 115	ANANYA 6.7%, ARJUN 6.7%, +13 more	20 – 22	CSE 33.3%, IT 26.7%, +3 more	7.85 – 9.75	HYDERABAD 33.3%, KHAMMAM 26.7%, +2 more
1	101	UDAY KIRAN	21	CSE	9.75	HYDERABAD
2	102	RAHUL	22	ECE	8.45	KHAMMAM
3	103	ANANYA	20	CSE	9.20	WARANGAL
4	104	SNEHA	21	IT	8.90	HYDERABAD
5	105	ARJUN	22	ECE	7.85	UNKNOWN

```
199 SELECT * FROM UDAY_COURSES;
200
201 SELECT * FROM UDAY_ENROLLMENTS;
202
203
204
```

Results (just now)

Table | Chart | Pivot | Next steps | 15 rows | 23ms

	COURSE_ID (301 – 315)	COURSE_NAME	CATEGORY	COURSE_FEE
		101 13.3% 102 13.3% +8 more	201 20.0% 205 20.0% +6 more	Completed 60.0% Enrolled 40.0%
11	311	110	208	Enrolled
12	312	102	202	Completed
13	313	103	206	Enrolled
14	314	107	201	Completed
15	315	109	203	Enrolled

```
201 SELECT * FROM UDAY_ENROLLMENTS;
202
203
204
```

Results (just now)

Table | Chart | Pivot | Next steps | 15 rows | 22ms

	ENROLLMENT_ID (301 – 315)	STUDENT_ID (101 – 110)	COURSE_ID (201 – 208)	ENROLLMENT_STATUS
				COMPLETED 60.0% ENROLLED 40.0%
1	301	101	201	COMPLETED
2	302	101	205	COMPLETED
3	303	102	203	COMPLETED
4	304	103	205	COMPLETED
5	305	104	202	ENROLLED

Feedback

Perform analytical queries

1. Total number of students

```
204 SELECT COUNT(*) AS TOTAL_STUDENTS
205 FROM UDAY_STUDENTS;
206
207
208
```

Results (just now)

Table | Chart | Pivot

	TOTAL_STUDENTS
1	15

Average CGPA

```sql
SELECT ROUND(AVG(CGPA), 2) AS AVERAGE_CGPA
FROM UDAY_STUDENTS;
```

Results (just now)

Table | Chart | Pivot

	AVERAGE_CGPA
1	8.90

Highest and lowest CGPA

```sql
SELECT
    MAX(CGPA) AS HIGHEST_CGPA,
    MIN(CGPA) AS LOWEST_CGPA
FROM UDAY_STUDENTS;
```

Results (just now)

Table | Chart | Pivot — 1 row — 40ms

	HIGHEST_CGPA	LOWEST_CGPA
1	9.75	7.85

Count students in each city

```sql
SELECT
    CITY,
    COUNT(*) AS TOTAL_STUDENTS
FROM UDAY_STUDENTS
GROUP BY CITY
ORDER BY TOTAL_STUDENTS DESC;
```

Results (just now)

Table | Chart | Pivot — 4 rows — 47ms

	CITY	TOTAL_STUDENTS
1	HYDERABAD	5
2	KHAMMAM	4
3	WARANGAL	4
4	UNKNOWN	2

Average CGPA by city

```sql
SELECT
    CITY,
    ROUND(AVG(CGPA), 2) AS AVERAGE_CGPA
FROM UDAY_STUDENTS
GROUP BY CITY
ORDER BY AVERAGE_CGPA DESC;
```

Results (just now)

Table | Chart | Pivot — 4 rows — 49ms

	CITY	AVERAGE_CGPA
1	WARANGAL	9.20
2	HYDERABAD	8.85
3	KHAMMAM	8.85
4	UNKNOWN	8.50

Perform JOIN operations

```
232
233  SELECT
234      S.STUDENT_ID,
235      S.STUDENT_NAME,
236      S.DEPARTMENT,
237      E.ENROLLMENT_STATUS
238  FROM UDAY_STUDENTS S
239  INNER JOIN UDAY_ENROLLMENTS E
240      ON S.STUDENT_ID = E.STUDENT_ID;
```

Results (just now)

Table | Chart | Pivot — 15 rows — 43ms

	STUDENT_ID	STUDENT_NAME	DEPARTMENT	ENROLLMENT_STATUS
	101 – 110	ANANYA 13.3% RAHUL 13.3% +8 more	CSE 40.0% ECE 20.0% +3 more	COMPLETED 60.0 ENROLLED 40.0
1	101	UDAY KIRAN	CSE	COMPLETED
2	101	UDAY KIRAN	CSE	COMPLETED
3	102	RAHUL	ECE	COMPLETED
4	102	RAHUL	ECE	COMPLETED
5	103	ANANYA	CSE	COMPLETED

Find the number of enrollments from each city

```
248  SELECT
249      S.CITY,
250      COUNT(E.ENROLLMENT_ID) AS TOTAL_ENROLLMENTS
251  FROM UDAY_STUDENTS S
252  INNER JOIN UDAY_ENROLLMENTS E
253      ON S.STUDENT_ID = E.STUDENT_ID
254  GROUP BY S.CITY
255  ORDER BY TOTAL_ENROLLMENTS DESC;
```

Results (just now)

Table | Chart | Pivot — 4 rows — 320ms

	CITY	TOTAL_ENROLLMENTS
1	HYDERABAD	7
2	WARANGAL	4
3	KHAMMAM	3
4	UNKNOWN	1

SQL Query Optimizations:

Create a table with the random records :

```
CREATE OR REPLACE TABLE UDAY_STUDENT_PERFORMANCE AS

SELECT
   SEQ4() + 1 AS RECORD_ID,

   UNIFORM(1000, 9999, RANDOM()) AS STUDENT_ID,

   GET(
      ARRAY_CONSTRUCT(
         'UDAY',
         'RAHUL',
         'ANANYA',
         'SNEHA',
         'ARJUN',
         'PRIYA',
         'KARTHIK',
         'MEGHANA',
         'VARUN',
         'DIVYA'
      ),
      UNIFORM(0, 10, RANDOM())
   )::VARCHAR AS STUDENT_NAME,

   GET(
      ARRAY_CONSTRUCT(
         'CSE',
         'ECE',
         'IT',
         'AIML',
         'MECHANICAL'
      ),
      UNIFORM(0, 5, RANDOM())
   )::VARCHAR AS DEPARTMENT,

   GET(
      ARRAY_CONSTRUCT(
         'HYDERABAD',
         'KHAMMAM',
         'WARANGAL',
         'VIJAYAWADA',
         'CHENNAI'
      ),
      UNIFORM(0, 5, RANDOM())
   )::VARCHAR AS CITY,
```

```
  UNIFORM(18, 26, RANDOM()) AS AGE,

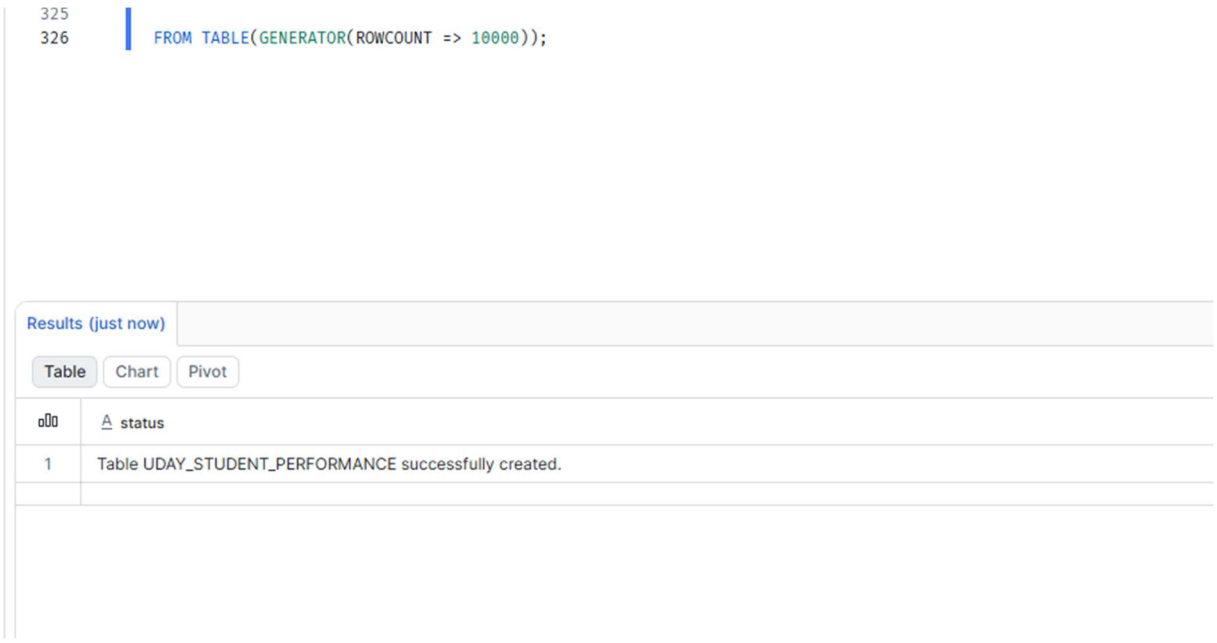
  ROUND(UNIFORM(600, 1000, RANDOM()) / 100.0, 2) AS CGPA,

  UNIFORM(3000, 15000, RANDOM()) AS COURSE_FEE,

  DATEADD(
     DAY,
     UNIFORM(0, 1095, RANDOM()),
     '2023-01-01'::DATE
  ) AS ENROLLMENT_DATE,

  GET(
     ARRAY_CONSTRUCT(
        'ENROLLED',
        'COMPLETED',
        'PENDING',
        'CANCELLED'
     ),
     UNIFORM(0, 4, RANDOM())
  )::VARCHAR AS ENROLLMENT_STATUS

FROM TABLE(GENERATOR(ROWCOUNT => 10000));
```



```
329    SELECT COUNT(*) AS TOTAL_RECORDS
330    FROM UDAY_STUDENT_PERFORMANCE;
```

Results (just now)

Table | Chart | Pivot

	TOTAL_RECORDS
1	10000

SELECT * vs selecting specific columns

```
333    SELECT *
334    FROM UDAY_STUDENT_PERFORMANCE;
```

Results (just now)

Table | Chart | Pivot — 10,000 rows — 400ms

	RECORD_ID	STUDENT_ID	STUDENT_NAME	DEPARTMENT	CITY	AGE	CGPA	COURSE_FEE	ENROLLMENT_DATE	ENROLLMENT_STATUS
1	1	1143	VARUN	AIML	KHAMMAM	20	7.03	12700	2024-11-06	PENDING
2	2	5051	null	AIML	CHENNAI	20	8.80	9294	2024-06-09	CANCELLED
3	3	3737	KARTHIK	CSE	KHAMMAM	23	6.26	9849	2025-04-05	null
4	4	1882	SNEHA	AIML	WARANGAL	26	8.65	5778	2024-02-20	PENDING
5	5	6357	SNEHA	MECHANICAL	VIJAYAWADA	21	9.28	5594	2025-12-02	COMPLETED

```
336    SELECT
337        STUDENT_NAME,
338        DEPARTMENT,
339        CGPA
340    FROM UDAY_STUDENT_PERFORMANCE;
```

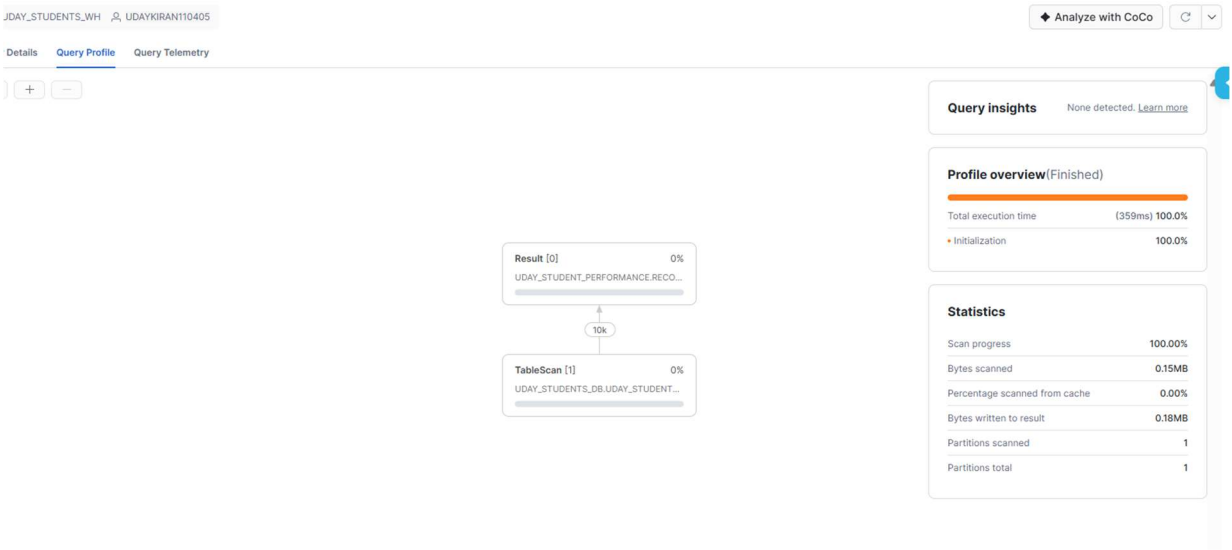
Results (just now)

Table | Chart | Pivot | Next steps — 10,000 rows — 113ms

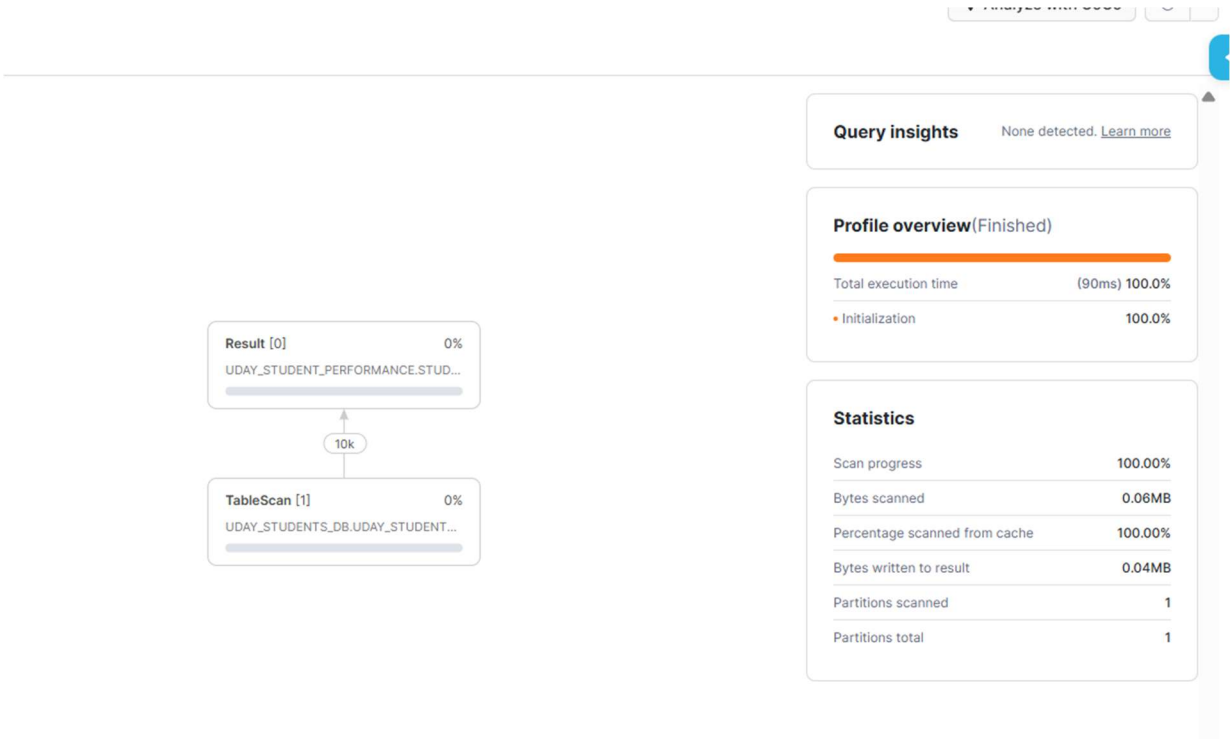
	STUDENT_NAME	DEPARTMENT	CGPA
1	VARUN	AIML	7.03
2	null	AIML	8.80
3	KARTHIK	CSE	6.26
4	SNEHA	AIML	8.65
5	SNEHA	MECHANICAL	9.28
6	DIVYA	AIML	8.94

Checking The Query Profile :

For all Records



For Selected Records :



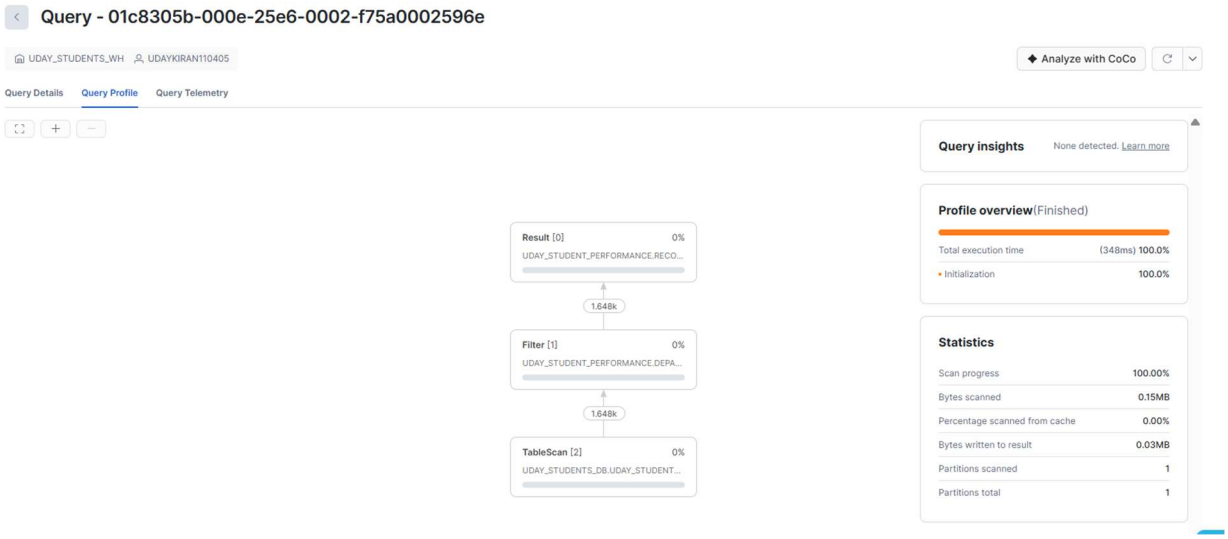
Apply WHERE conditions

```
342  SELECT *
343  FROM UDAY_STUDENT_PERFORMANCE
344  WHERE DEPARTMENT = 'CSE';
```

Results (just now)

Table | Chart | Pivot | Next steps — 1,648 rows — 388ms

	RECORD_ID	STUDENT_ID	STUDENT_NAME	DEPARTMENT	CITY	AGE	CGPA	COURSE_FEE	ENROLLMENT_DATE	ENROLLMENT_STATUS
1	3	3737	KARTHIK	CSE	KHAMMAM	23	6.26	9849	2025-04-05	null
2	38	6046	RAHUL	CSE	WARANGAL	24	8.78	12451	2024-09-09	PENDING
3	40	6456	KARTHIK	CSE	CHENNAI	24	7.70	12071	2024-09-08	null
4	43	9534	MEGHANA	CSE	CHENNAI	26	9.24	5416	2024-09-07	COMPLETED
5	51	8314	SNEHA	CSE	KHAMMAM	24	9.56	6260	2023-05-22	CANCELLED
6	52	1507	RAHUL	CSE	null	26	9.00	11764	2023-03-08	null



Date-based filtering

Fetch records from a specific date range

```
347  SELECT
348      STUDENT_NAME,
349      ENROLLMENT_DATE,
350      CGPA
351  FROM UDAY_STUDENT_PERFORMANCE
352  WHERE ENROLLMENT_DATE BETWEEN
353      '2024-01-01' AND '2024-12-31';
```

Results (just now)

Table | Chart | Pivot — 3,289 rows — 307ms

	STUDENT_NAME	ENROLLMENT_DATE	CGPA
	KARTHIK 10.4% UDAY 10.3% +8 more	1/1/2024 – 12/31/2024	6 – 10
1	VARUN	2024-11-06	7.03
2	null	2024-06-09	8.80
3	SNEHA	2024-02-20	8.65
4	VARUN	2024-05-09	9.66

Feedback

Fetch recent records

```
355     SELECT
356     STUDENT_NAME,
357     ENROLLMENT_DATE,
358     DEPARTMENT
359 FROM UDAY_STUDENT_PERFORMANCE
360 WHERE ENROLLMENT_DATE >= '2025-01-01';
```

Results (just now)

Table | Chart | Pivot

3,387 rows | 53ms

	STUDENT_NAME	ENROLLMENT_DATE	DEPARTMENT
	DIVYA 9.7% PRIYA 9.6% +8 more	1/1/2025 – 12/31/2025	MECHANICAL 18.2% IT 16.4% +3 more
1	KARTHIK	2025-04-05	CSE
2	SNEHA	2025-12-02	MECHANICAL
3	PRIYA	2025-08-22	AIML
4	RAHUL	2025-04-04	null

Comparing the Query Profile:

Query insights None detected. Learn more

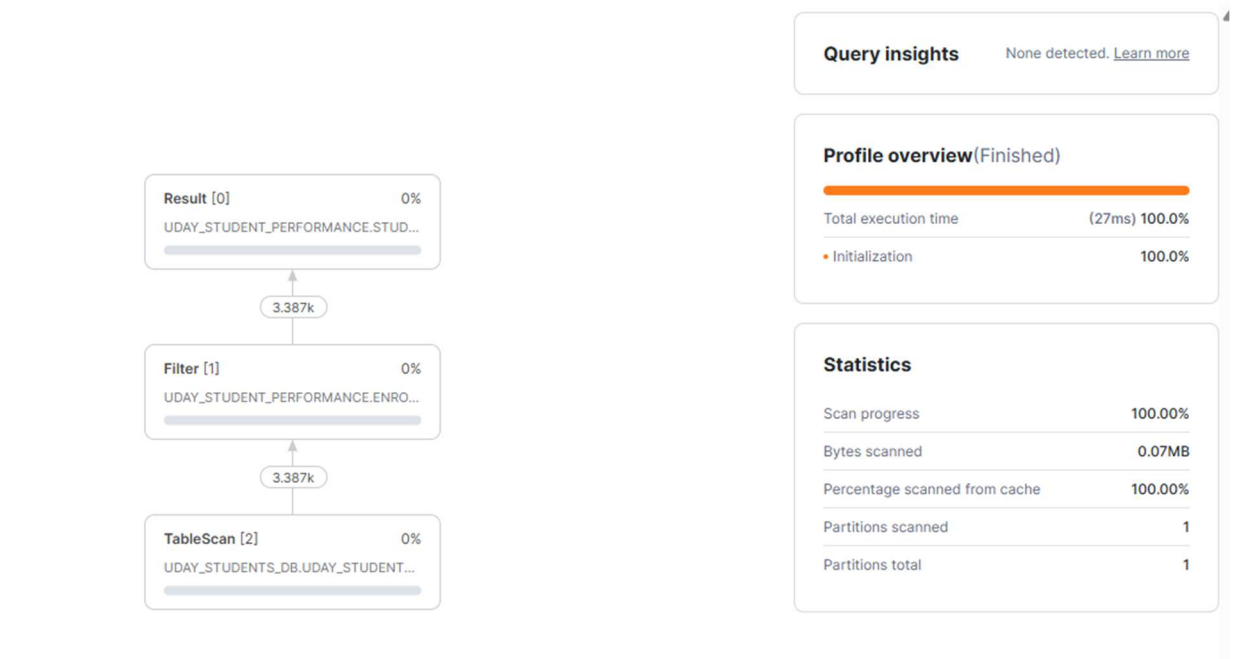
Profile overview (Finished)

Total execution time	(281ms) 100.0%
• Initialization	100.0%

Statistics

Scan progress	100.00%
Bytes scanned	0.15MB
Percentage scanned from cache	0.00%
Partitions scanned	1
Partitions total	1

For The total one



Took all the data from cache only

Reason:

Concept: Micro-partition Pruning

Snowflake stores metadata about micro-partitions, including minimum and maximum values.

When you filter by a date, Snowflake can skip partitions whose date ranges cannot match your condition.